

Compiling Written Payment Policy into Governed, Executable Rules

A Generative AI Opportunity for Cotiviti

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Defining the Concept

Content management in health care covers the creation, maintenance, and governance of the documents that decide how care is paid for. These include billing and coding policies, clinical practice guidelines, and payer to provider contracts. Within that domain one task stays stubbornly manual: the conversion of written policy into executable logic. A payment policy is authored in prose for a human to read. Before a claims system can act on it, an analyst has to translate that prose into a precise rule, a feature, or a model that software can run. This report treats that translation step as its subject and names it policy compilation, the act of turning a written payment or coding policy into a transparent rule that a deterministic engine can apply to claims, with every rule traceable back to the exact clause that produced it.

The distinction that matters is between authoring a rule and applying one. Applying rules to claims is a mature discipline. Rule engines have run claim edits for decades, and this capability is the heart of Cotiviti's Payment Policy Management product, which helps payers tailor, test, and execute clinical and payment policies so that incorrect payments are averted (Cotiviti, 2024). Authoring the rule from the written policy, by contrast, is still done by clinicians and certified coders working line by line. Generative AI now makes that authoring step a candidate for assistance for the first time, because a capable model can read prose and emit structured output that a schema validates.

The Problem This Solves

A large national health plan maintains thousands of payment policies, and those policies change constantly as code sets, clinical evidence, and regulations move. Each change starts a manual chain. A subject matter expert rereads the revised policy, decides what the change means for the edit, rewrites the logic, and tests it. That chain is slow, expensive, and inconsistent from one author to the next. The lag between a policy update and the matching rule update is a direct source of incorrect payments, the precise outcome the payment integrity industry exists to prevent. Cotiviti serves more than one hundred payers, including twenty three of the top twenty five health plans, and stakes its reputation on administering policy correctly at a measured accuracy of 99.90 percent (Cotiviti, 2024). The constraint on that operation is not the engine that applies the rules but the human effort required to author and maintain them.

Policy compilation attacks that constraint directly. A working demonstrator built for this assessment, PolicyForge, shows the full loop. An analyst pastes a written policy. A model compiles it into a typed rule that states a logic type, the target codes, the conditions that fire, any exceptions, an outcome, and a confidence score, and every condition carries the verbatim phrase from the policy that justifies it. A deterministic engine then applies that rule to claims and returns pay, flag, or deny for each one, with the firing condition and the cited clause attached. The design rests on one principle. The model authors the rule and a deterministic engine applies it. That separation keeps every adjudication reproducible and auditable, because the same claim always yields the same result once the rule is fixed. Any rule whose confidence falls below a set threshold, along with boundary cases, routes to a human review queue rather than deciding on its own. The result is assistance that drafts and explains, not an automated system that decides without oversight.

Relevant Trends

Several trends make this the right moment for the capability. Generative models have become reliable at structured generation, so prose can be turned into validated, typed output rather than free text, which is the precondition for safe rule authoring. Cost pressure is intense. McKinsey estimates that generative AI could automate a meaningful share of health plan administrative work and reduce medical costs by

improving claim review, edits, and recovery, a savings opportunity measured in hundreds of billions of dollars (McKinsey & Company, 2023a, 2023b). Adoption is accelerating, with Gartner forecasting that most organizations will run generative AI models in production within a short horizon (Gartner, 2023). Regulators and buyers now expect transparency and explainability, which favors approaches that cite their evidence and keep a human accountable. Early commercial activity is also concentrated on the provider and revenue cycle side, while the payer side task of authoring the governed rule itself remains largely open.

Opportunities and Threats

The opportunities are concrete. Policy compilation can shorten the time from a published policy to a working edit from days to minutes, which reduces the window in which incorrect payments occur. It can raise consistency, because the same policy produces the same draft rule regardless of which analyst handles it. It can attach a citation and a confidence score to every edit, which strengthens the audit trail and supports the accuracy story Cotiviti already tells. It can also reduce provider abrasion, since a transparent rule with a plain language reason is easier to defend than an opaque denial.

The threats are equally concrete. A model can produce a rule that is plausible but wrong, it can encode bias if the source material carries it, and either failure can create regulatory exposure. There is also a strategic threat. The hand built rule library is a core asset, and a tool that drafts rules automatically could appear to commoditize it. The mitigation for each is the same architecture the demonstrator uses. Keep a human in the loop with a confidence gate, apply rules with a deterministic engine so behavior is reproducible, and require a verbatim citation for every condition so each rule can be checked against its source. Framed this way, the technology defends the asset rather than erodes it, because it makes expert judgment faster and better documented rather than replacing it.

Strategic Options for Cotiviti

The PolicyForge demonstrator built for this assessment is the tangible proof behind the options below, and the same artifact supplies the material for the slides and the video. Three options follow, in priority order.

Option one. A governed policy authoring assistant inside Payment Policy Management. Build a copilot that drafts the executable rule from policy text, presents it to a certified expert for approval, cites the exact clause behind every condition, and routes low confidence drafts to review. This is the primary recommendation, because it sits on the existing flagship product, it matches the stated philosophy that AI is a tool that augments expert coders rather than replacing them, and it addresses the maintenance burden that Cotiviti's own research leadership has named as a generative AI priority.

Option two. A policy change intelligence service. Monitor policy updates, summarize what materially changed in plain language, recompile only the affected rule, and present a clause by clause comparison of the old and new versions. This serves the maintenance pain directly and turns a reactive manual task into a monitored, documented workflow.

Option three. A traceability and audit layer. Attach a citation and a confidence score to every edit across the portfolio, regardless of how it was authored, to harden the responsible AI posture and make the accuracy claim independently checkable.

The recommended path is to pursue option one as a supervised copilot and to add option two as its natural extension, while keeping the deterministic engine as the system of record so that adjudication stays reproducible. This captures the speed and consistency gains, holds an expert accountable for every rule, and converts a perceived commoditization threat into a durable speed advantage for the people who already own the work.

Conclusion

Applying rules to claims is the mature part of payment integrity. Authoring those rules from written policy is the slow and costly part, and that is where generative AI can help today without surrendering control. A governed compiler that cites its evidence and keeps an expert accountable, shown end to end by PolicyForge, is a measured and commercially relevant position for Cotiviti to pursue.

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